Viswa Sai Ammiraju Bonam

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Skills

Programming & Query Languages: Python, SQL, Java, R
Cloud & Data Platforms: AWS, Azure (Microsoft Certified: Azure Data Scientist Associate)
Big Data & Other: Apache Kafka, Apache Spark, DBT, Data Modeling, ETL/ELT Pipeline Design, Data Orchestration
Databases: MySQL, PostgreSQL, MongoDB, DynamoDB, NoSQL, FAISS
Machine Learning & Analytics: Feature Engineering, Predictive Modeling, SHAP Analysis, Model Tuning, Data Preprocessing
Visualization Tools: Power BI, Tableau, Streamlit, Matplotlib, Seaborn

Professional Experience

Data Analyst Jul 2024 – Aug 2024
UTDSOM Investment Corporation *Richardson, United States*

- Engineered a **data processing pipeline** using **Python** to **ingest, parse,** and **clean** unstructured data from financial statements and SEC filings, which served as the foundation for an ML model that improved prediction accuracy by 20%.
- Developed a **data model** to process and rank stock performance based on expected returns, creating a system that improved portfolio returns by 10%.
- Implemented **data validation** and **partitioning logic** using **Pandas** and **NumPy** to create reliable estimation and forecast datasets, enhancing the stability of the data model by 15%.

Associate Database Developer Jul 2022 – Aug 2023
SkyEss Techno Solutions *Hyderabad, India*

- Engineered and optimized relational **database schemas** in **MySQL** using advanced data modeling techniques, which accelerated query performance and reduced data retrieval time by 30%.
- Built and maintained **data integration pipelines** using **Python** and **Pandas** to serve backend data to front-end interfaces, improving data access efficiency for the guest management system by 25%.
- Automated data quality checks and validation routines in **SQL** to ensure **data consistency** across business reporting, which increased overall reporting accuracy by 40%.

Projects

End-to-End E-Commerce Data Platform on Azure

- Architected and deployed a scalable end-to-end ETL/ELT pipeline on **Azure**, ingesting data from 3+ disparate sources (**MySQL, MongoDB**, and flat files) into **Azure Data Lake Storage Gen2.**
- Orchestrated data workflows with **Azure Data Factory** and implemented a multi-layered **Medallion architecture**, leveraging **Azure Databricks and Apache Spark** to transform raw data into cleansed and aggregated datasets, improving data quality and reliability by 99.9%.
- Engineered and optimized complex **data transformations** in **Python** and **Spark**, reducing data processing time by 30% and preparing the data for the **serving layer (Azure Synapse)** to power business intelligence and data science initiatives.

American Express Credit Default Prediction (UT Dallas)

- Built an ML tool to predict credit default using **XGBoost**, **Logistic Regression** and **Neural Network model** by applying **hyperparameter tuning** and **feature engineering**, which resulted in a 94.7% AUC score.
- Engineered key variables using **Python** and **Pandas**, reducing overfitting and improving model stability.
- Implemented credit scoring strategies using advanced **statistical analysis**, reducing default rates to 5% and 10%, enhancing decision-making and profitability.

Equity Insight: AI-Driven News Analysis Tool (Personal)

- Built an automated news research tool using **LangChain**, **OpenAI**, and **Python**, reducing manual article review time by 60%, and accelerating equity analysis insights.
- Utilized **FAISS vector database** to optimize **semantic search** across articles, improving query accuracy by 35%.
- Deployed **Streamlit** UI that enhanced user experience, delivering real-time insights and boosting productivity by 50% for equity analysts.

Predictive Modeling with Multivariate Time-Series Features (UT Dallas)

- Developed a predictive model using XGBoost to forecast XRP price returns, improving classification accuracy by **15%** - methodology applicable to **user retention or content performance predictions.**
- Applied **SHAP analysis** for model interpretability, surfacing top predictive features and reducing noise by **40%** - showcasing a deep understanding of model transparency for business insights.
- Integrated **multi-variable datasets** (social sentiment, asset behavior) to simulate real-world signals - approach relevant to modeling **streaming engagement signals** like watch time, content category, and release timing.

Education

The University of Texas at Dallas Richardson, Texas
Master of Science in Financial Technology and Analytics *Aug 2023 - May 2025*
JNTUH College of Engineering Hyderabad, India
Bachelor of Technology in Mechanical Engineering *Aug 2018 - Jun 2022*